

Superintelligent Retrieval Agent: The Next Frontier of Agentic Retrieval

Zeyu Yang^{1,2,*}, Qi Ma¹, Jason Chen¹, Anshumali Shrivastava^{1,2,*}

¹Meta Superintelligence Labs, ²Rice University

*Work done at Meta

Retrieval-augmented agents are increasingly the interface to large organizational and public knowledge bases, yet most still treat retrieval as a black box: they issue exploratory queries, inspect returned snippets, and iteratively reformulate until useful evidence emerges. This resembles how a newcomer searches an unfamiliar database rather than how an expert navigates it with strong priors about terminology, constraints, and likely evidence, leading to unnecessary retrieval rounds, increased latency, and poor recall.

We introduce *Superintelligent Retrieval Agent* (SIRA), which defines *superintelligence* in retrieval as the ability to compress multi-round exploratory search into a single corpus-discriminative retrieval action. SIRA does not merely ask what terms are relevant to the query; it asks which terms are likely to separate the desired evidence from corpus-level confusers. On the corpus side, an LLM enriches each document offline with missing search vocabulary; on the query side, it predicts evidence vocabulary omitted by the query; and corpus statistics are used as tool calls to filter proposed terms that are absent, overly common, or unlikely to create retrieval margin. The final retrieval step is a single weighted BM25 call combining the original query with the validated expansion.

Across ten BEIR benchmarks, SIRA achieves the strongest average retrieval performance in our comparison, outperforming dense retrievers, learned sparse retrievers, and LLM-based search-agent baselines while using no relevance labels or retriever fine-tuning. On downstream question answering, SIRA’s retrieval-only answer coverage exceeds recent RL-trained agentic QA systems on NQ and HotpotQA. Finally, we introduce **BrowseComp-Wikipedia**, a hard-search benchmark of 232 BrowseComp-derived queries grounded in a 25,587,229-document English Wikipedia index. Even without index-time LLM document enrichment, using only grounded Wikipedia categories as corpus-visible structure, SIRA outperforms multi-round Perplexity agents at every retrieval budget, reaching 9.70% Recall@1, 15.27% Recall@10, and 36.14% Recall@100. These results show that one well-formed, corpus-grounded lexical retrieval action can outperform substantially more expensive multi-round search while remaining interpretable, training-free, and efficient.

Date: June 8, 2026

Correspondence: Zeyu Yang at zy45@meta.com and Anshumali Shrivastava anshumali@meta.com

Code: Available at <https://github.com/facebookresearch/sira>

BrowseComp-Wikipedia Dataset: Available at <https://github.com/facebookresearch/sira>



1 Introduction

Information retrieval (IR) has evolved from lexical matching, exemplified by BM25 (Robertson and Zaragoza, 2009), to neural retrieval dominated by dense embeddings (Karpukhin et al., 2020). Embedding-based retrievers perform well when trained with abundant in-domain supervision: large-scale relevance labels and interaction logs that calibrate the model to a platform’s user population (Bajaj et al., 2016; Joachims et al., 2007; Chapelle and Zhang, 2009). This regime has fueled modern *retrieval-augmented generation* (RAG) systems that ground LLM outputs in external corpora (Lewis et al., 2020).

The user interface for information access is changing rapidly: search is increasingly *answer-forward* and

conversational, with LLMs mediating multi-turn information seeking (Mo et al., 2025). A key consequence is that the classic training signal for supervised ranking, clickthrough, becomes sparse, delayed, and biased: users often terminate sessions without clicking, or accept an on-page summary as the final answer. A large-scale browsing analysis by Pew Research Center finds that when Google presents an AI-generated summary, users click standard result links substantially less often (Pew Research Center, 2025). Click-based supervision is therefore becoming unreliable at scale, precisely as query behavior is shifting.

Compositional queries need controllable retrieval. At the same time, *query distributions* are moving away from short keyword strings toward longer, compositional requests that combine constraints, exclusions, and multi-step intent. This shift is a hallmark of conversational search. Pure similarity search is an awkward fit for this regime: dense retrieval exposes only a black-box nearest-neighbor operator and provides weak handles for enforcing structure (e.g., must-include/must-not-include constraints, attribute filters, or explicit decomposition). Neural sparse methods such as SPLADE (Formal et al., 2021) partially restore lexical controllability while preserving learning-based ranking, but they are still used as fixed retrievers inside pipelines rather than as controllable components of an agent policy.

Classical lexical retrieval, exemplified by BM25, possesses underappreciated strengths that become decisive when paired with LLM reasoning. BM25 is *transparent*: an agent can boost keywords, enforce constraints, and decompose queries with predictable effects on retrieval outcomes. It naturally rewards *rare*, *discriminative terms* via IDF weighting, so domain-specific jargon that would be diluted in a dense embedding becomes a powerful retrieval signal. It is *auditable*: one can trace exactly which keywords matched and why, while avoiding the latency and memory costs of dense indices. The missing ingredient has been a mechanism to surface the right rare terms and constraints; LLMs, with their vast parametric knowledge, are uniquely positioned to fill this role.

LLM reasoning meets retrieval. The limitations of dense retrieval are not merely engineering inconveniences; they expose a deeper mismatch between single-vector representations and compositional information needs. Recent theoretical and empirical work shows that fixed-dimensional embeddings can realize only a limited family of relevance patterns (Weller et al., 2025), that static embeddings are information bottlenecks whenever relevance requires cross-attention-style interaction (Shrivastava, 2025), and that vector databases impose substantial cost, latency, and objective-mismatch burdens in production (Shrivastava, 2023)

In parallel, LLM reasoning frameworks such as Chain-of-Thought, Tree-of-Thoughts, and Graph-of-Thoughts demonstrate that LLMs can plan and explore structured intermediate states (Wei et al., 2022; Yao et al., 2023; Besta et al., 2024). Tool-using agents extend this capability to external actions, including search (Yao et al., 2022), and recent reinforcement-learning approaches train LLMs to interleave reasoning with multi-turn web search (Jin et al., 2025). In most agentic search systems, however, retrieval itself remains an opaque tool: the agent can rewrite queries and judge snippets, but cannot directly manipulate retrieval primitives such as keyword weighting, constraint selection, or decomposition tied to index-time signals.

This exposes why current search agents remain brittle despite strong reasoning capabilities. Agentic systems such as ReAct, IRCoT, and recent RL-trained search agents improve performance by interleaving reasoning with repeated retrieval calls (Yao et al., 2022; Trivedi et al., 2023; Jin et al., 2025). However, this success is partly obtained through a *retrieval-context advantage*: after each search, the agent absorbs returned snippets, discovered entities, surface vocabulary, and near misses into the LLM context, then uses this accumulated evidence to formulate later queries. In other words, the agent compensates for weak retrieval control by learning the corpus through interaction. This strategy is expensive and noisy, and it relies increasingly on long-context LLMs to retain and use many intermediate passages—a regime known to be unreliable when relevant evidence is buried in long contexts (Liu et al., 2024). Thus the failure mode is not simply that LLMs cannot reason; it is that the search interface gives them too little direct control, forcing them into hit-and-miss exploration.

The known hardness of retrieval that LLM agents are clueless about. Classical IR theory clarifies the missing ingredient. Retrieval is not just a question of whether a query is semantically related to the desired document; it is a comparative ranking problem in which the gold evidence must outrank many non-gold *confusers*. The probability ranking principle and learning-to-rank methods formalize this as ordering relevant

documents above non-relevant alternatives (Robertson, 1977; Joachims et al., 2007; Chapelle and Zhang, 2009), while BM25 operationalizes corpus contrast through document frequency and IDF (Robertson and Zaragoza, 2009). A query can therefore be plausible in isolation yet fail because its terms also match many distractors, or because its most expert-sounding terms are absent, too common, or weakly discriminative in the target index. Dense single-vector retrieval adds a further bottleneck: recent theory shows that fixed-dimensional embedding retrievers cannot realize all top- k relevance patterns and can fail even on simple realistic constraint structures (Weller et al., 2025). This points to the core gap SIRA addresses: an LLM may know what relevant evidence should look like, but it needs index-visible statistics and explicit retrieval controls to make that expectation discriminative against the confusing articles that corpus may be flooded with.

SIRA formalizes the setting of one-shot, controllable BM25 retrieval across broad IR and QA benchmarks, building on recent evidence that LLM agents can generate discriminative keyword, grep, and ripgrep queries approaching RAG-level QA performance (Subramanian et al., 2025; Wang et al., 2026). Existing results, however, are either practitioner-level, code-centric, or multi-turn and context-accumulating (Karpathy, 2026; Pan et al., 2025); SIRA targets the stronger single-query regime.

A particularly important stress test is hard web-style search. BrowseComp was introduced to measure whether browsing agents can locate hard-to-find, short-answer facts on the web (Wei et al., 2025). Its questions are valuable because they are difficult to find but easy to verify, but the original benchmark relies on live web search, where the corpus, ranking function, and snippets are uncontrolled. BrowseComp-Plus improves reproducibility by constructing a fixed corpus of 100,195 documents for 830 verified queries (Chen et al., 2025). However, a curated 10^5 -document corpus is still much smaller than the search spaces faced by real agents, and it weakens the core disambiguation problem: ranking the right page above millions of plausible confusers. We therefore introduce BrowseComp-Wikipedia, a 232-query benchmark derived from BrowseComp and grounded entirely in a 25,587,229-document English Wikipedia index. This setting preserves the hard-search character of BrowseComp while making retrieval controlled, public-corpus grounded, and substantially larger scale.

Goal. The central limitation of today’s LLM-driven retrieval agents is that retrieval remains a *black-box environment* they explore through repeated interaction. Existing agents issue a query, inspect returned snippets, and reformulate using accumulated evidence, a *retrieval-context advantage* where later searches are conditioned on information exposed by earlier ones. This resembles a domain newcomer learning an unfamiliar database through exploration, rather than an expert who anticipates relevant terminology and evidence patterns before reading retrieved passages, as illustrated in Figure 1.

We define *superintelligence in retrieval* as the ability to replace this multi-round process with a single expert-level retrieval action: (i) form a domain-informed expectation of what relevant evidence looks like, (ii) ground that expectation using lightweight index-aware signals (document frequency), (iii) compile the result into explicit retrieval controls, and (iv) execute retrieval efficiently and transparently.

1.1 Our Proposal: SIRA

We propose the *Superintelligent Retrieval Agent* (SIRA), a retrieval-centric agent that searches like a domain expert. Modern LLMs encode substantial parametric knowledge such as terminology, entities, relations, but lack a mechanism to convert this knowledge into precise retrieval actions over a target corpus. SIRA provides that mechanism through a scalable two-stage framework.

First, the LLM produces an *expected-response sketch*: a compact hypothesis of the concepts, entities, and discriminative terms likely to appear in relevant evidence. This sketch acts as a retrieval prior, not as evidence. Before issuing the final BM25 query, SIRA consults lightweight corpus-statistics tools (document frequencies) to validate and prune proposed terms without returning answer passages, avoiding the retrieval-context advantage enjoyed by multi-round agents.

Second, conditioned on the sketch and index statistics, SIRA compiles a *retrieval program*: a single controlled BM25 query with weighted keywords, optional exclusions, and structured composition. We evaluate SIRA in this strict one-shot setting: one LLM reasoning step, optional index-statistic checks, and one BM25 call. Multiple queries could improve performance, but the one-query regime isolates the central question: how far can retrieval go when the agent must formulate the right lexical action without reading retrieved snippets?

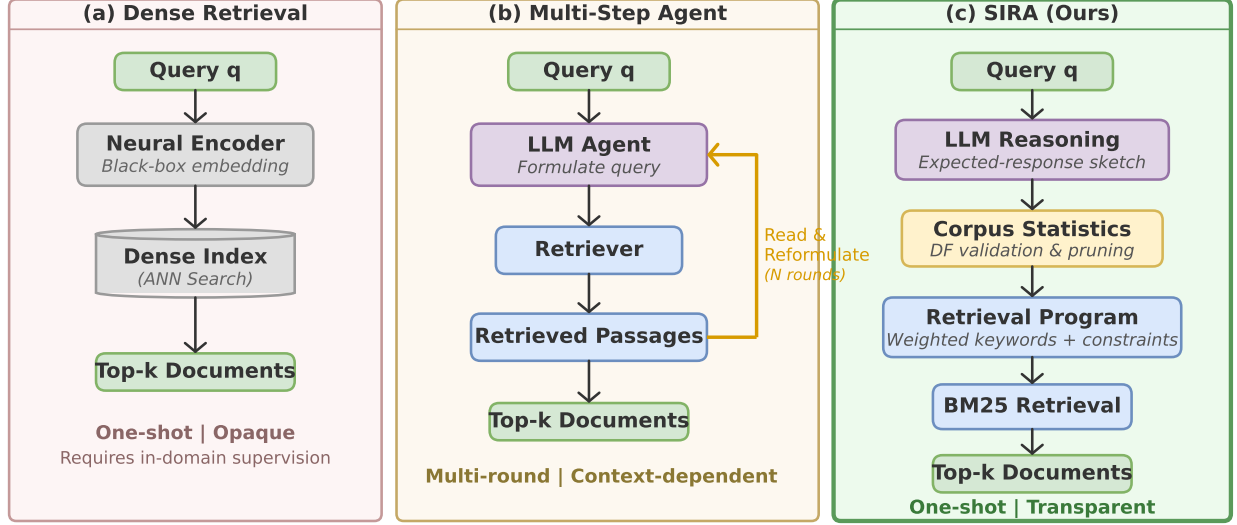


Figure 1 Three retrieval paradigms compared. **(a)** Dense retrieval encodes queries and documents into a shared embedding space and performs nearest-neighbor search; the process is one-shot but opaque and requires in-domain supervision. **(b)** Multi-step agent retrieval uses an LLM to iteratively formulate queries, read retrieved passages, and reformulate over N rounds; later queries benefit from accumulated retrieval context. **(c)** SIRA produces an expert-level retrieval action in a single shot: the LLM generates an expected-response sketch, validates proposed terms against corpus statistics, and compiles a controlled BM25 query with weighted keywords and constraints, all without reading any retrieved passages.

BM25’s reliance on exact matching becomes a strength when the LLM supplies the right vocabulary and verifies its discriminative value through corpus statistics.

We evaluate SIRA in three increasingly demanding settings. First, across ten BEIR-style retrieval benchmarks (Thakur et al., 2021), SIRA outperforms strong dense retrievers, learned sparse retrievers, and LLM-based retrieval-agent baselines on average Recall@10 and NDCG@10. Second, on downstream QA, SIRA’s retrieval-only answer coverage exceeds recent RL-trained agentic QA systems on NQ and HotpotQA, even though those systems use trained search policies and answer generators. Third, on BrowseComp-Wikipedia, SIRA directly challenges the prevailing multi-round search-agent paradigm: a single corpus-grounded BM25 query, constructed before reading retrieved pages, outperforms Perplexity-based agents that can browse iteratively for up to 100 turns. These results suggest that the bottleneck in retrieval-augmented agents is not the sophistication of the reader or the number of search iterations, but the agent’s ability to construct an expert-level, corpus-discriminative retrieval action.

2 Background and Organization

BM25 and sparse lexical retrieval. BM25 (Robertson and Zaragoza, 2009) is the dominant lexical ranking function in modern search systems. It scores a document d against a query $q = (q_1, \dots, q_n)$ by summing per-term contributions:

$$\text{BM25}(q, d) = \sum_{i=1}^n \underbrace{\log \left(1 + \frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} \right)}_{\text{IDF}(q_i)} \cdot \frac{f(q_i, d)}{f(q_i, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}} \right)}, \quad (1)$$

where $f(q_i, d)$ is the frequency of term q_i in d , $|d|$ is the document length in tokens, avgdl is the average document length across the corpus, N is the corpus size, and $n(q_i)$ is the number of documents containing q_i .

The formula decomposes into two interpretable factors. We use the Lucene variant, which applies $\log(1 + x)$ to the classical Robertson–Spärck Jones ratio, guaranteeing non-negative IDF for all terms. The IDF term

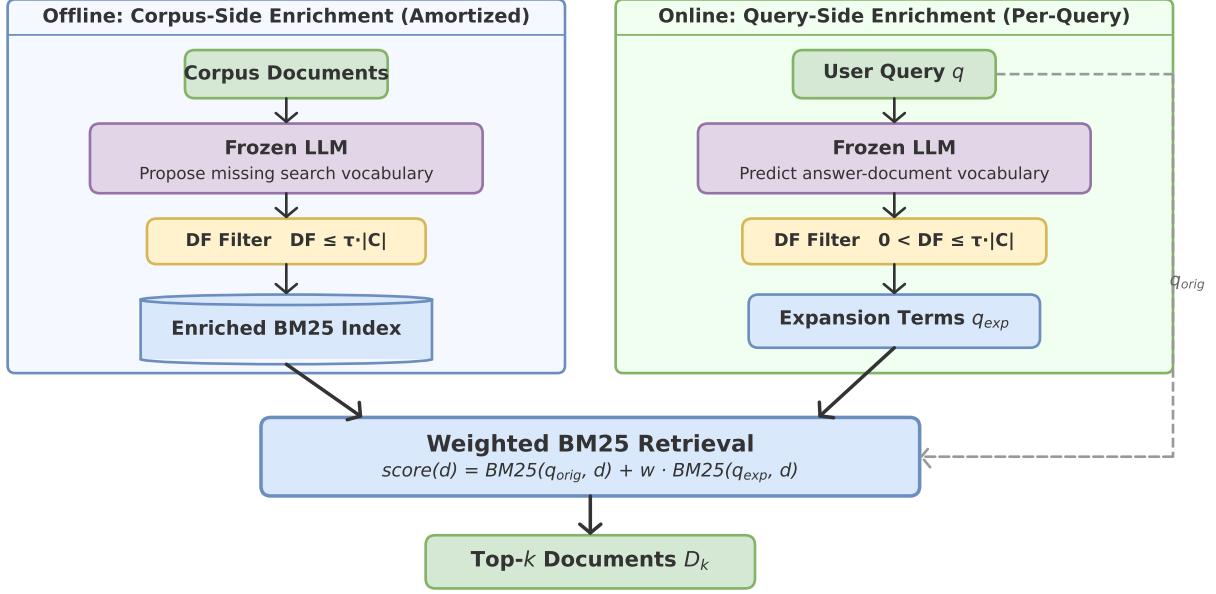


Figure 2 SIRA pipeline overview. Corpus-side enrichment (left) is performed once offline; query-side enrichment (right) runs per query. Both stages apply a DF filter to reject uninformative terms; the query-side filter additionally requires $DF > 0$ to ensure each expansion term exists in the index. The original query q_{orig} bypasses enrichment (dashed) and is combined with the expansion terms in a single weighted BM25 call.

down-weights common words and up-weights rare, discriminative terms: a query term appearing in most documents contributes near-zero IDF, while one appearing in only a handful of documents receives a large weight. The TF saturation term ensures that repeating a word within a document yields diminishing returns, controlled by k_1 . The parameter b governs length normalization: at $b=1$ the score fully normalizes for document length, while at $b=0$ document length is ignored.

BM25 is implemented over an inverted index, which maps each vocabulary term to the documents and term frequencies in which it appears.

Index-visible signals. BM25 makes corpus contrast observable through the inverted index: before retrieval, an agent can check whether a candidate term appears in the corpus, how many documents contain it, and how much IDF weight it can contribute. These signals do not reveal answer passages, but they expose whether LLM-proposed vocabulary is absent, too common, or likely to create retrieval margin. SIRA uses this information to validate, prune, and weight expansion terms so the final query is not merely plausible in isolation, but discriminative within the target corpus.

3 SIRA: Superintelligent Retrieval Agent

3.1 Overview

Most retrieval-augmented agents interact with a search engine through a loop: issue a query, inspect results, reformulate, and repeat until useful evidence emerges. SIRA replaces this multi-round loop with a *one-shot* pipeline that bridges the vocabulary gap between queries and documents from *both sides* simultaneously. The full system requires no training, no relevance labels, and no supervised query–document pairs; it operates with a frozen LLM and corpus statistics alone.

How SIRA expertizes a new corpus. Given an unseen corpus, SIRA builds domain expertise from both the corpus side and the query side, as illustrated in Figure 2. On the *corpus side* (offline, once per corpus),

the LLM reads each document, anticipates the search vocabulary a user would need to find it, and proposes candidate terms absent from the document text. A document-frequency filter validates each candidate against the corpus index, discarding uninformative terms whose frequency exceeds an upper bound (detailed in Section 3.2). The surviving terms are injected into the BM25 index as atomic n-gram entries. On the *query side* (online, per query), the LLM first produces an *expected-response sketch*: a compact set of concepts, entities, and discriminative terms likely to appear in a relevant document but absent from the query. The same DF filter grounds the sketch in the enriched index. Conditioned on the validated sketch, SIRA then compiles a *retrieval program* and executes a single weighted BM25 call (detailed in Section 3.2):

$$\text{score}(d) = \text{BM25}(q_{\text{orig}}, d) + w \cdot \text{BM25}(q_{\text{exp}}, d), \quad (2)$$

where q_{orig} is the original query, q_{exp} is the filtered expansion, and w is the expansion weight. After the BM25 retrieval top- k candidates are picked from the returned candidates based on relevance with the query.

3.2 Vocabulary Enrichment with Superintelligent LLMs

SIRA bridges the query–document vocabulary gap by enriching both sides with LLM-proposed terms. The goal of enrichment is not to add more text indiscriminately, but to surface compact lexical signals: concepts, aliases, and phrases that are likely to identify the desired evidence while remaining rare enough to distinguish it from the rest of the corpus.

This requires grounding LLM proposals in corpus statistics. A term that sounds expert-like may be useless for query-side enrichment if it never appears in the enriched index, and weak if it appears across many unrelated documents. SIRA therefore treats document frequency and BM25/TF-IDF-style salience as lightweight corpus-statistic tools: they check query-side term existence, estimate whether proposed terms are too common to discriminate, and retain terms that can contribute meaningful retrieval margin.

DF filter. To ensure that only corpus-grounded and discriminative terms enter the retrieval pipeline, both enrichment stages share a *document-frequency (DF) filter*. The filter enforces an upper bound $\text{DF} \leq \tau \cdot |C|$, pruning terms that are repeated across too much of the corpus and therefore receive little useful IDF weight. For query-side enrichment, the filter additionally requires $\text{DF} > 0$, ensuring that every expansion phrase actually exists in the enriched index and can affect BM25 scoring. Corpus-side enrichment does not require this lower bound, since enrichment itself introduces new vocabulary into the index. The result is a compact set of terms that are plausible under the LLM’s domain knowledge and measurable as useful search signals in the target corpus.

Corpus-side enrichment (offline). The goal is to anticipate how a user would search for a document when the vocabulary they would use is absent from the document text. The prompt explicitly instructs the LLM to generate *new* terms not already present in the document, focusing on discriminative vocabulary: synonyms, abbreviations, alternate names, and domain-specific phrasings that maximize lexical contrast with existing index terms. Crucially, the prompt is task-aware: for claim-verification corpora, it emphasizes entity aliases and factual cues; for argument retrieval, it targets opposing-side vocabulary; for duplicate detection, it focuses on intent-preserving synonym substitutions. Phrases that pass the DF filter are decomposed into sliding-window n-grams and injected into the corpus index as additional posting-list entries.

Query-side enrichment (online). The goal is the mirror image of corpus-side enrichment: predict vocabulary that a relevant answer document would use but that is absent from the query. The prompt instructs the LLM to generate discriminative *topic and domain vocabulary* that narrows the search space, while explicitly forbidding it from guessing the answer itself. This distinction is critical for factoid queries, where predicting a named entity (e.g., a person or date) would bias retrieval toward a single candidate rather than broadening coverage of relevant evidence. As on the corpus side, the prompt is task-aware: factoid QA targets contextual terms surrounding the answer, multi-hop QA distributes expansion across all entities and reasoning hops, and duplicate detection focuses on intent-preserving synonym substitutions.

4 Experiments

We evaluate SIRA in three stages. First, we test pure retrieval quality on ten BEIR benchmarks. Second, we ask whether this retrieval advantage transfers to downstream question answering. Third, we evaluate hard search-agent retrieval on BrowseComp-Wikipedia, a new BrowseComp-derived benchmark grounded in a 25,587,229-document English Wikipedia index.

4.1 Experimental Setup

We evaluate SIRA on ten BEIR datasets (Thakur et al., 2021) spanning seven retrieval task types: question answering (NQ, HotpotQA), opinion retrieval (FIQA), fact-checking (FEVER, Climate-FEVER, SciFact), argument retrieval (ArguAna), citation prediction (SciDocs), and duplicate detection (Quora, CQADupStack). Table 1 summarizes corpus sizes (5K to 5.4M documents) and the number of relevant documents per query (1.0 to 29.9), covering a broad range of retrieval regimes. We report Recall@10 and NDCG@10. Recall@10 measures the fraction of relevant documents appearing in the top-10 results, directly capturing a system’s ability to surface evidence for a downstream reader. NDCG@10 additionally rewards placing relevant documents at higher ranks, providing a complementary view of ranking quality.

BEIR is a particularly stringent testbed for retrieval agents because it evaluates retrieval directly rather than end-to-end answer generation. Many recent agentic search systems focus on QA-style benchmarks where a reader model, prior knowledge, or multi-round interaction can partially mask weak retrieval. In contrast, BEIR exposes the core retrieval problem: given a fixed corpus and a query, can the system surface the relevant documents in the top ranks? This makes it the right setting for evaluating SIRA as a retrieval agent rather than as a general QA agent.

Table 1 Overview of the ten BEIR retrieval benchmarks used in our evaluation, spanning diverse reasoning types including fact-checking, argumentation, citation prediction, and standard QA. The suite covers a broad range of domains and corpus sizes (5K to 5.4M documents), rigorously testing generalization. **Rel D/Q** denotes the average number of relevant documents per query.

Dataset	Type	Queries	Corpus	Rel D/Q	Description
NQ	Question Answering	3,452	2.68M	1.22	Retrieval for real-world Google search questions.
HotpotQA	Question Answering	7,405	5.23M	2.00	Multi-hop reasoning over Wikipedia paragraphs.
FIQA	Opinion Retrieval	648	57K	2.63	Financial QA over StackExchange data.
ArguAna	Argument Retrieval	1,401	8.67K	1.00	Matching counter-arguments for debate topics.
CQADupStack	Duplicate Question	1,570	40K	2.40	Duplicate detection across StackExchange forums.
Quora	Duplicate Question	10,000	523K	1.57	Duplicate detection for Quora questions.
SciDocs	Citation Prediction	1,000	26K	29.93	Predicting citations for scientific papers.
FEVER	Fact-Checking	6,666	5.42M	1.19	Verifying claims against Wikipedia text.
Climate-FEVER	Fact-Checking	1,535	5.42M	3.05	Verifying climate change claims.
SciFact	Fact-Checking	300	5K	1.13	Verifying scientific claims against abstracts.

Baselines. We compare against ten baselines spanning three retrieval paradigms. BM25 (Robertson and Zaragoza, 2009) is the standard sparse lexical baseline. Among neural methods, E5 (Wang et al., 2022) is a dense bi-encoder trained on large-scale relevance data; SPLADE (Formal et al., 2021) and SPARTA (Zhao et al., 2021) are learned sparse retrieval models that preserve inverted-index efficiency while learning term importance weights; Doc2Query (Nogueira et al., 2019) performs document expansion by predicting queries a document would answer. Among LLM-based methods, HyDE (Gao et al., 2023) generates a hypothetical document as a BM25 query expansion; CoT (Wei et al., 2022) uses chain-of-thought prompting to expand the query with reasoning-derived terms; Search-R1 (Jin et al., 2025) trains an RL policy for multi-round search; GrepRAG (Wang et al., 2026) generates grep-like pattern queries originally designed for code retrieval; and ShellAgent (Subramanian et al., 2025) relies on grep-based keyword-search tools within an agentic loop for multi-step retrieval. HyDE, CoT, GrepRAG, ShellAgent, and SIRA use the same frozen LLM (Qwen3.6-35B-A3B-FP8); Search-R1 (E5) uses its publicly available trained checkpoint with the most potent E5 backend. Neural baselines use official checkpoints (see Table 2 caption). SIRA uses Qwen3.6-35B-A3B-FP8 as its frozen LLM for both corpus-side and query-side enrichment.

4.2 Main Results on Large-Scale Information Retrieval

In Table 2, we report Recall@10 and NDCG@10 across all ten benchmarks, comparing SIRA against sparse, dense, and agentic baselines. We structure the analysis around three research questions:

- **RQ1:** Can a training-free agentic system match or surpass trained sparse and dense retrievers?
- **RQ2:** Why do generic LLM search agents lag behind retrieval-native systems on BEIR, and does SIRA close this gap?
- **RQ3:** Does SIRA’s retrieval quality translate to downstream QA performance competitive with RL-trained agentic QA systems?

Table 2 evaluates the central claim of this paper: a frozen LLM can turn BM25 from a lexical baseline into the strongest retriever in the comparison. SIRA uses the LLM not as a reader and not merely as a query-expansion module, but as a controller for the BM25 engine itself. It proposes missing vocabulary, grounds those proposals with corpus statistics, weights the surviving terms through BM25’s IDF-sensitive scoring surface, and executes the result as a single ranked retrieval call.

This comparison is deliberately strict. The baselines include supervised dense and sparse retrievers trained on large-scale relevance data, as well as LLM-based query-expansion and search-agent baselines. Thus, the key question is not whether LLMs can generate plausible search text or perform more search rounds, but whether they can produce a corpus-discriminative retrieval action that ranks gold evidence above confusers.

Table 2 Recall@10 and NDCG@10 on ten BEIR datasets. HyDE, CoT, GrepRAG, ShellAgent, and SIRA use Qwen3.6-35B-A3B-FP8 (frozen, 3B active parameters); Search-R1 (E5) uses its publicly available checkpoint with an E5 retrieval backend. Neural baselines use official checkpoints: SPARTA (BeIR/sparta-msmarco-distilbert-base-v1), SPLADE (naver/splade-cocondenser-ensembledistil), Doc2Query (doc2query/msmarco-t5-base-v1), E5 (intfloat/e5-base-v2). Best per dataset in **bold**; second best underlined.

	ArguAna	C-FEVER	CQADup	FEVER	FIQA	HotpotQA	NQ	Quora	SciDocs	SciFact	Avg
<i>Recall@10</i>											
BM25	.7738	.1764	.4163	.6747	.3198	.6141	.4543	.9014	.1636	.8078	.5302
Doc2Query	.7824	.1761	.4339	.6769	.3397	.6527	.5068	.8975	.1663	.8270	.5459
SPARTA	.6181	.1050	.3100	.7246	.2450	.5356	.5584	.7445	.1303	.7084	.4680
SPLADE	<u>.8137</u>	.2881	.4924	.8954	.4139	.7027	.7381	.9206	.1654	.8230	.6253
E5	.7909	.2899	<u>.5138</u>	<u>.9109</u>	<u>.4697</u>	<u>.7276</u>	.7877	.9428	.1962	<u>.8489</u>	<u>.6478</u>
HyDE	.7091	.2598	.3299	.7132	.2845	.5051	.4918	.5151	.1530	.8344	.4796
CoT	.7752	.1867	.4020	.6765	.3086	.5789	.4961	.8704	.1595	.7961	.5250
Search-R1 (E5)	.5760	<u>.3014</u>	.5008	.9010	.4499	.6705	.7889	.9355	<u>.2018</u>	.8349	.6161
GrepRAG	.5746	.0176	.2400	.1635	.1009	.3434	.1628	.5105	.1027	.5883	.2804
ShellAgent	.2263	.0305	.2084	.2557	.1035	.4327	.1884	.3330	.0843	.6685	.2531
SIRA	.9036	.3025	.6301	.9114	.4904	.7536	<u>.7883</u>	<u>.9390</u>	.2676	.9216	.6908
<i>NDCG@10</i>											
BM25	.4874	.1372	.3481	.5036	.2532	.5851	.2916	.8055	.1565	.6791	.4247
Doc2Query	.4946	.1381	.3667	.5122	.2731	.6299	.3302	.7960	.1589	.6920	.4392
SPARTA	.3890	.0852	.2497	.6101	.1925	.5132	.3983	.6294	.1272	.5894	.3784
SPLADE	.5253	.2293	.4083	.7933	.3478	.6869	.5369	.8344	.1586	.7025	.5223
E5	<u>.5323</u>	<u>.2397</u>	<u>.4196</u>	<u>.8096</u>	.3932	.6905	<u>.5835</u>	.8648	.1855	<u>.7156</u>	<u>.5434</u>
HyDE	.4366	.2004	.2463	.5507	.2223	.4451	.3315	.3924	.1402	.6565	.3622
CoT	.4951	.1471	.3354	.4932	.2486	.5595	.3168	.7608	.1518	.6647	.4173
Search-R1 (E5)	.3658	.2654	.4040	.8215	.3765	.6520	.5790	<u>.8543</u>	<u>.1878</u>	.7094	.5216
GrepRAG	.3555	.0122	.2048	.0971	.0763	.2927	.0908	.4557	.0921	.4129	.2090
ShellAgent	.1114	.0206	.1558	.1298	.0686	.3417	.1036	.2476	.0727	.4374	.1689
SIRA	.6174	.2288	.5327	.8037	<u>.3771</u>	<u>.6904</u>	.5923	.8490	.2449	.7866	.5723

RQ1: SIRA surpasses trained retrievers without supervision. SIRA achieves the highest average Recall@10 on BEIR, reaching 0.691 compared with 0.648 for E5, 0.625 for SPLADE, and 0.530 for BM25. It does so without relevance labels, without fine-tuning a retriever, and without building an embedding index. The advantage also holds for ranking quality: SIRA reaches 0.572 average NDCG@10, compared with 0.543 for E5 and 0.522 for SPLADE. This is the central result: a training-free retrieval agent built on BM25 can outperform supervised dense and learned sparse retrievers trained on large-scale relevance data.

The gains are broad rather than driven by a single dataset. SIRA obtains the best Recall@10 on eight of ten benchmarks; the two exceptions are NQ, where Search-R1 (E5) is ahead by 0.06 percentage points, and Quora, where E5 is ahead by 0.4 percentage points. The largest improvements over E5 appear on datasets with structural query–document vocabulary gaps: +36% relative on SciDocs, +23% on CQADupStack, and +14% on ArguAna. These are precisely the settings where corpus-grounded enrichment should help: the LLM proposes missing terminology, the DF filter removes absent or overly common terms, and BM25 amplifies the surviving discriminative vocabulary through IDF-weighted scoring.

RQ2: SIRA turns LLM reasoning into retrieval-native ranking. The LLM-based baselines remain below SIRA on pure retrieval metrics, but Search-R1 (E5) clarifies an important distinction. HyDE and CoT remain close to BM25 on average, with Recall@10 of 0.480 and 0.525 compared with 0.530 for BM25. Search-R1 gains substantially from its E5 backend, reaching 0.616 Recall@10 and 0.522 NDCG@10, but still trails SIRA’s 0.691 Recall@10 and 0.572 NDCG@10. Stronger backend retrieval clearly helps, but a multi-round search policy on top of that backend still does not close the gap to corpus-grounded BM25 control.

The gap is even larger for grep-style tool-use agents. GrepRAG and ShellAgent use the same LLM backbone as SIRA, but treat retrieval as pattern generation rather than corpus-aware ranking, yielding average Recall@10 of only 0.280 and 0.253. Because the backbone is shared, the gap isolates the retrieval interface: grep-style agents search with patterns that lack BM25’s document-frequency and IDF-weighted term scoring, while SIRA turns LLM proposals into weighted retrieval signals. SIRA therefore outperforms GrepRAG and ShellAgent by 41.0 and 43.8 absolute Recall@10 points.

Recent LLM search agents also show how much performance depends on the retrieval backend: Search-R1 improves markedly when paired with E5, yet still remains below E5 on average Recall@10 and below SIRA on both Recall@10 and NDCG@10. This does not mean E5 reasons better; it means retrieval-native systems are optimized for the object BEIR measures: ranking relevant documents above corpus-level distractors. SIRA closes this gap by using the LLM to program the retrieval engine itself: proposed terms are grounded by corpus statistics, filtered for discriminative value, weighted through BM25’s IDF-sensitive scoring surface, and executed in a single ranked retrieval call.

4.3 Downstream Question Answering

Our goal is to test whether SIRA’s retrieval advantage translates into downstream question answering. Since SIRA is a retrieval agent rather than an answer generator, the evaluation requires QA datasets with an associated retrieval corpus and gold evidence/answer annotations. Among the BEIR benchmarks, this leaves NQ and HotpotQA: both provide fixed corpora for retrieval and gold answer strings that allow us to measure whether retrieved documents contain the evidence needed to answer.

This setting directly tests the claim that a better retrieval agent can become a better QA agent by supplying stronger evidence to the reader. It is also a difficult comparison for SIRA. The baselines are recent agentic QA systems designed for these tasks, and many are trained or reinforced on QA-style objectives closely aligned with NQ and HotpotQA. They report end-to-end generated-answer accuracy, so they may receive credit even when retrieval is incomplete, because the reader can use parametric knowledge or answer synthesis. In contrast, SIRA is counted correct only when the gold answer string appears in the retrieved text.

We compare against six recent RL-trained agentic QA systems, including Search-R1 (Jin et al., 2025), TIPS (Xie et al., 2026), A²Search (Zhang et al., 2025), E-GRPO (Zhang et al., 2026), SSP (Lu et al., 2025), and HiPRAG (Wu et al., 2025). We take the best reported numbers directly from the original papers, giving each baseline the benefit of its full end-to-end QA pipeline. SIRA uses no reader model in this comparison. It also uses no RL fine-tuning, no task-specific supervised training, and no multi-round search; the same

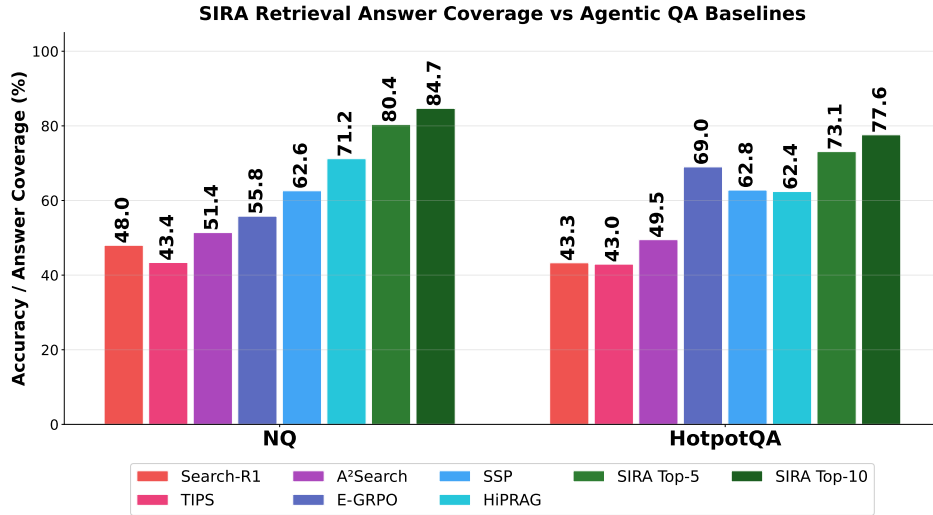


Figure 3 SIRA retrieval answer coverage (top-5 and top-10) vs. six RL-trained agentic QA systems on NQ and HotpotQA. Baseline numbers are the best reported results taken directly from each original paper. All baselines are end-to-end QA pipelines reporting generated-answer accuracy; SIRA is a pure retriever with no reader. Answer coverage requires the gold string to be retrieved.

one-shot SIRA retriever used in the BEIR experiments is evaluated directly. SIRA is evaluated only by *answer coverage*, the fraction of queries for which its retrieved documents contain verifiable answer evidence.

RQ3: The best retrieval agent yields the strongest QA evidence. Despite this disadvantaged evaluation, SIRA’s retrieval-only answer coverage exceeds the reported end-to-end QA accuracy of all six agentic baselines at top-10. This comparison gives the baselines both their trained search policies and their answer generators, while SIRA contributes only retrieved text. As shown in Figure 3, SIRA reaches 84.7% on NQ and 77.6% on HotpotQA. The strongest baselines: HiPRAG reaches 71.2% on NQ, while E-GRPO reaches 69.0% on HotpotQA. SIRA exceeds both by large margins.

The result is also strong at a tighter retrieval budget. At top-5, SIRA achieves 80.4% on NQ and 73.1% on HotpotQA, still exceeding every baseline in the comparison. These results support a simple conclusion: for corpus-grounded QA, improving retrieval can be more important than adding more search rounds or training the answer generator. The compared agents are built for end-to-end QA and many are optimized on QA-style rewards, yet SIRA surfaces answer-bearing passages more reliably with a single retrieval call.

4.4 BrowseComp-Wikipedia: Hard Search at Wikipedia Scale

BrowseComp was introduced as a benchmark for evaluating browsing agents on hard-to-find, short-answer questions that require nontrivial search over the web (Wei et al., 2025). The benchmark is valuable because the answers are easy to verify once found, while the evidence-finding process can require substantial exploration. However, for evaluating search itself, the original BrowseComp setting has two important limitations. First, it depends on live web search, so the searchable corpus, ranking function, snippets, and availability of evidence are controlled by an external and changing system. Second, as BrowseComp has become a widely used benchmark for deep-research and browsing agents, the questions and answers are increasingly exposed through papers, leaderboards, logs, and repeated evaluations, making leakage and overfitting a growing concern.

A recent line of work has tried to make BrowseComp-style evaluation more reproducible by converting it into a fixed-corpus setting. BrowseComp-Plus, for example, contains 830 queries over a fixed corpus of 100,195 documents (Chen et al., 2025). This is an important step toward controlled retrieval evaluation, but the corpus is still small relative to the setting in which search agents are normally expected to operate. In a corpus of roughly one hundred thousand documents, the search problem can be made easier by benchmark-specific engineering, memorization, or overfitting to a relatively limited evidence pool. This is precisely the issue we argue against throughout the paper: the central difficulty of search is not merely producing a semantically plausible query, but disambiguating the right page from a very large number of plausible confusers. That difficulty becomes much more realistic when the search space contains millions of pages rather than only a small curated corpus.

To create a harder and more realistic evaluation, we construct **BrowseComp-Wikipedia**, a new benchmark that combines BrowseComp-style query difficulty with Wikipedia-scale retrieval. We identify 232 BrowseComp queries whose answers can be fully supported using English Wikipedia pages alone. For each query, we annotate the gold Wikipedia URL or URLs that contain sufficient evidence to answer the question. The retrieval corpus is an indexed English Wikipedia collection containing **25,587,229 documents**. This creates a substantially larger and more natural retrieval setting than existing fixed-corpus BrowseComp variants: the queries retain the hard, indirect, entity-seeking structure of BrowseComp, but the search system must recover the answer-bearing page from a corpus with tens of millions of Wikipedia documents.

This benchmark fills an important gap in search-agent evaluation. Original BrowseComp evaluates end-to-end browsing over the live web. BrowseComp-Plus provides a controlled but relatively small fixed-corpus benchmark. BrowseComp-Wikipedia instead evaluates whether a search agent can retrieve the correct Wikipedia page from a large, public, reproducible corpus. The task is deliberately strict: success requires retrieving the gold page that can support the answer, not merely generating the correct final answer from parametric memory or from indirectly related evidence.

SIRA without index-time LLM enrichment. For this experiment, we evaluate a deliberately toned-down version of SIRA. In the main method, SIRA uses index-time enrichment to add missing search vocabulary to documents. Here, we remove that expensive LLM-based document enrichment step. Instead, we use Wikipedia’s own category system as a natural source of corpus-side enrichment. Each Wikipedia document is indexed using its text, unigrams, bigrams, and category strings. Category strings are treated as protected lexical features: for example, a category such as `1927 births` is indexed as an atomic feature such as `category:1927_births`, rather than being split into unrelated terms.

This category information is useful because many BrowseComp-style questions describe the target entity indirectly. A question may not name the answer directly, but it may imply constraints such as a birth year, death year, occupation, nationality, award, genre, institution, historical period, or place. Wikipedia categories expose many of these attributes in a corpus-visible form. Examples include categories such as `1927 births`, `American film actresses`, `British science fiction writers`, `Nobel laureates in Physics`, and `People educated at Eton College`. These categories give SIRA precise retrieval handles for attributes that may be only indirectly expressed in the original query.

To prevent the LLM from hallucinating category constraints, SIRA is not allowed to freely invent category tokens. During query construction, the LLM may propose potentially useful Wikipedia categories, but every proposed category must be grounded through the Wikipedia category graph before it can be used in the final BM25 query. The category-graph tool verifies whether the category exists and can return valid neighboring categories when appropriate. Invalid or unsupported categories are discarded. Thus, the LLM contributes semantic search planning, but the final retrieval query remains grounded in corpus-observable structure.

For BrowseComp-Wikipedia, SIRA is not single-query unlike in the easier strict BEIR sense. Instead, it is evidence-blind and budgeted: the agent may issue multiple category-grounded BM25 calls, but every returned page consumes the fixed retrieval budget, and recall is measured over the resulting stream of unique Wikipedia pages. SIRA may fetch titles/snippets only for pages already inside the fixed page budget, and fetched content is used only for final ranking, not for formulating subsequent BM25 queries.

Agentic search with the Perplexity search engine. As strong agentic baselines, we equip each frozen backbone LLM with the same production web-search engine and allow it to browse iteratively. Concretely, the agent is given a single retrieval tool, `perplexity_search_wikipedia_en`, which performs a Perplexity-based web search and returns a ranked list of up to 20 results per query. Each result is returned as structured output containing the title, URL, and snippet, which the agent can read before deciding what to do next. To match the evidence space available to SIRA, the search tool is constrained to `en.wikipedia.org`: all other pages, including non-English Wikipedia subdomains, are filtered out. This gives the agent a clean, single-source evidence stream from English Wikipedia.

The Perplexity baseline follows the standard BrowseComp evaluation protocol with the default prompt. The agent can issue searches, inspect returned titles and snippets, reformulate its query, and continue browsing until it commits to an answer, for up to 100 turns per question. This baseline represents the prevailing

multi-round search-agent paradigm: an LLM controls a commercial search engine through repeated query reformulation and intermediate result inspection. In contrast, SIRA represents a corpus-grounded retrieval paradigm: the LLM must compile a retrieval query against the indexed corpus before seeing retrieved pages.

Evaluation protocol. Our evaluation protocol is intentionally strict for SIRA. The LLM is allowed to think as much as needed before issuing a BM25 query, and it may use the category-graph tool to validate candidate category constraints. The prompt used for SIRA is shown in Figure 4. However, once the BM25 query is fired, every returned page consumes retrieval budget. SIRA is not allowed to inspect retrieved snippets or pages and then reformulate the query using that retrieved context. This differs from the Perplexity baseline, which can read intermediate search results and use them to issue better follow-up queries.

We evaluate retrieval budgets of $B \in \{1, 10, 100\}$ pages. A query is counted as successful if at least one gold Wikipedia URL appears among the pages returned within the budget. For SIRA, this corresponds directly to the top B BM25 results. For the Perplexity baseline, we consider the ordered stream of unique English Wikipedia URLs surfaced to the agent during its iterative browsing process and truncate that stream at the same budgets. This allows us to measure gold-page recall under comparable evidence budgets, even though the baseline is allowed to browse interactively.

The resulting metric is gold-page recall: the fraction of the 232 queries for which at least one annotated gold Wikipedia page is retrieved within the page budget. Formally,

$$\text{Recall@}B = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}[G_q \cap R_B(q) \neq \emptyset],$$

where $\mathcal{Q}$ is the set of BrowseComp-Wikipedia queries, G_q is the set of gold Wikipedia URLs for query q , and $R_B(q)$ is the set of at most B retrieved Wikipedia URLs. Recall@1 measures whether the system ranks the answer-bearing page first. Recall@10 measures whether the answer page fits within a small reader context. Recall@100 measures whether the retriever can recover the target page under a larger but still finite retrieval budget.

Results. The results are summarized in Table 3. The strongest SIRA configuration achieves the best recall at every retrieval budget, despite operating under a substantially stricter protocol than the agentic search baselines built on the Perplexity search engine. With Claude 4.6 Opus as the backbone, SIRA obtains 9.70% Recall@1, 15.27% Recall@10, and 36.14% Recall@100. Compared with the best Perplexity baseline at each budget, this corresponds to gains of 6.68 percentage points at Recall@1, 8.37 percentage points at Recall@10, and 3.81 percentage points at Recall@100.

The gains are largest in the low-budget regime, which is the most important setting for retrieval-augmented agents. Recall@1 measures whether the system places the answer-bearing page first, and Recall@10 measures whether the page is available within a small reader context. SIRA more than triples the best Perplexity Recall@1 score and more than doubles the best Perplexity Recall@10 score. This is notable because the Perplexity-search agents are allowed to browse interactively for up to 100 turns, reading titles and snippets before issuing follow-up searches. SIRA, by contrast, must commit to a single corpus-grounded BM25 query before reading any retrieved page.

These results support the central hypothesis of this paper: hard search is not only about having a powerful LLM or a strong commercial search engine, but about constructing the right corpus-grounded retrieval action. The Perplexity-search agents represent the prevailing multi-round paradigm, where an LLM repeatedly searches, reads partial results, and reformulates. SIRA instead uses the LLM to compile a precise retrieval program before evidence consumption. Its strong low-budget recall shows that Wikipedia categories provide discriminative, corpus-visible handles that allow SIRA to rank the correct page above millions of plausible confusers.

Overall, BrowseComp-Wikipedia provides a new controlled benchmark for hard search-agent evaluation. It preserves the difficulty of BrowseComp-style queries, but grounds evaluation in a public Wikipedia corpus with 25,587,229 indexed documents and gold URL supervision. It also separates two search paradigms: iterative web-search agents that rely on reading intermediate results, and corpus-grounded retrieval agents that must

System	Backbone LLM	Recall@1	Recall@10	Recall@100
<i>Agentic baseline: Perplexity search over English Wikipedia, up to 100 turns</i>				
Perplexity	Claude 4.6 Opus	<u>2.59</u>	<u>4.74</u>	<u>32.33</u>
Perplexity	GPT-5.4	<u>3.02</u>	<u>6.90</u>	31.47
<i>SIRA: single-shot, category-grounded BM25 on Wikipedia</i>				
SIRA	Claude 4.6 Opus	9.70	15.27	36.14
SIRA	GPT-5.4	5.71	13.13	<u>18.51</u>

Table 3 Gold-page recall on BrowseComp-Wikipedia. All values are percentages. The benchmark contains 232 BrowseComp queries whose answers can be fully supported using English Wikipedia. The retrieval corpus contains 25,587,229 indexed Wikipedia documents. SIRA uses no index-time LLM enrichment in this experiment; instead, it uses Wikipedia categories as natural corpus-side enrichment and validates proposed category tokens against the Wikipedia category graph. The Perplexity baseline browses interactively over en.wikipedia.org for up to 100 turns, while SIRA must issue corpus-grounded BM25 queries before reading any retrieved page. For each backbone LLM, the better of the two search engines (Perplexity vs. SIRA) is shown in **bold** and the other is underlined.

construct a precise search action before consuming evidence. The strong performance of SIRA in this setting supports the central claim of the paper: hard search is not only about semantic understanding, but about constructing grounded retrieval programs that can disambiguate the right page from a very large corpus.

Limitations and future work. SIRA assumes that the frozen LLM can understand the query and provide useful semantic priors about the target corpus. We have not evaluated settings where the corpus is far outside the LLM’s pretraining distribution; in such domains, corpus-side adaptation or fine-tuning may be needed before the LLM can propose reliable enrichment terms.

5 Conclusion

We introduced SIRA, a retrieval-centric agent that turns LLM reasoning into controllable lexical retrieval. Instead of using the LLM to repeatedly query a black-box search tool, read snippets, and reformulate, SIRA uses the LLM to program the retrieval action itself: enrich the corpus with missing user vocabulary, enrich the query with likely evidence vocabulary, validate proposed terms with corpus statistics, and execute a single weighted BM25 call.

The main result is that this simple interface changes the role of BM25. Across ten BEIR benchmarks, SIRA achieves the highest average Recall@10 and NDCG@10 in our comparison, outperforming BM25, E5, SPLADE, and recent LLM-based search agents while using no relevance labels, no retriever fine-tuning, and no embedding index. Its gains are broad, with the best Recall@10 on eight of ten datasets and especially large improvements on tasks where query and document vocabularies diverge.

The downstream QA results show that this retrieval advantage matters beyond standard IR metrics. On NQ and HotpotQA, SIRA’s retrieval-only answer coverage exceeds recent RL-trained agentic QA systems, even though those systems use trained search policies and answer generators while SIRA contributes only retrieved evidence. This supports a simple conclusion: for corpus-grounded QA, the ability to surface the right evidence can matter more than adding more search rounds.

Finally, BrowseComp-Wikipedia shows that the same principle extends to hard agentic search at much larger scale. We construct a 232-query benchmark from BrowseComp whose answers can be supported entirely by English Wikipedia, and evaluate retrieval over a 25,587,229-document Wikipedia index. In this setting, we remove SIRA’s index-time LLM document enrichment and use only Wikipedia categories as natural corpus-side structure, grounding every proposed category through the Wikipedia category graph. Even under this stripped-down protocol, SIRA outperforms Perplexity-based agents at every retrieval budget, despite the baselines being allowed to browse iteratively for up to 100 turns. The strongest SIRA configuration reaches 9.70% Recall@1, 15.27% Recall@10, and 36.14% Recall@100.

Together, these results suggest a different path for retrieval-augmented agents. Rather than making agents search longer, accumulate more context, and rely on increasingly expensive interaction with black-box search engines, we can make the retrieval action itself more expert, corpus-aware, and interpretable. The remaining open question is how far this idea extends to corpora far outside the frozen LLM’s knowledge, where corpus adaptation or fine-tuning may be needed before reliable enrichment is possible. But at least for broad public knowledge corpora and standard retrieval benchmarks, SIRA shows that one well-grounded retrieval program can beat many rounds of exploratory search.

References

- Payal Bajaj, Daniel Campos, Nick Craswell, Li Deng, Jianfeng Gao, Xiaodong Liu, Rangan Majumder, Andrew McNamara, Bhaskar Mitra, Tri Nguyen, et al. Ms marco: A human generated machine reading comprehension dataset. *arXiv preprint arXiv:1611.09268*, 2016.
- Maciej Besta, Nils Blach, Ales Kubicek, Robert Gerstenberger, Michal Podstawski, Lukas Gianinazzi, Joanna Gajda, Tomasz Lehmann, Hubert Niewiadomski, Piotr Nyczyk, et al. Graph of thoughts: Solving elaborate problems with large language models. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 17682–17690, 2024.
- Olivier Chapelle and Ya Zhang. A dynamic bayesian network click model for web search ranking. In *Proceedings of the 18th international conference on World wide web*, pages 1–10, 2009.
- Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, et al. Browsecmp-plus: A more fair and transparent evaluation benchmark of deep-research agent. *arXiv preprint arXiv:2508.06600*, 2025.
- Thibault Formal, Benjamin Piwowarski, and Stéphane Clinchant. Splade: Sparse lexical and expansion model for first stage ranking. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2288–2292, 2021.
- Luyu Gao, Xueguang Ma, Jimmy Lin, and Jamie Callan. Precise zero-shot dense retrieval without relevance labels. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1762–1777, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.99. <https://aclanthology.org/2023.acl-long.99/>.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Serkan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. Search-r1: Training llms to reason and leverage search engines with reinforcement learning. *arXiv preprint arXiv:2503.09516*, 2025.
- Thorsten Joachims, Laura Granka, Bing Pan, Helene Hembrooke, Filip Radlinski, and Geri Gay. Evaluating the accuracy of implicit feedback from clicks and query reformulations in web search. *ACM Transactions on Information Systems (TOIS)*, 25(2):7–es, 2007.
- Andrej Karpathy. LLM wiki. GitHub Gist, <https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f>, 2026. Accessed: 2026-05-05.
- Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 6769–6781, 2020.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems*, 33:9459–9474, 2020.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranajape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638. <https://aclanthology.org/2024.tacl-1.9/>.
- Hongliang Lu, Yuhang Wen, Pengyu Cheng, Ruijin Ding, Jiaqi Guo, Haotian Xu, Chutian Wang, Haonan Chen, Xiaoxi Jiang, and Guanjun Jiang. Search self-play: Pushing the frontier of agent capability without supervision. *arXiv preprint arXiv:2510.18821*, 2025.
- Fengran Mo, Kelong Mao, Ziliang Zhao, Hongjin Qian, Haonan Chen, Yiruo Cheng, Xiaoxi Li, Yutao Zhu, Zhicheng Dou, and Jian-Yun Nie. A survey of conversational search. *ACM Transactions on Information Systems*, 43(6):1–50, 2025.
- Rodrigo Nogueira, Wei Yang, Jimmy Lin, and Kyunghyun Cho. Document expansion by query prediction. *arXiv preprint arXiv:1904.08375*, 2019.
- Ben Pan, Carlo Baronio, Albert Tam, Pietro Marsella, Mokshit Jain, Daniel Chiu, Swyx, and Silas Alberti. Introducing swe-grep and swe-grep-mini: RL for multi-turn, fast context retrieval. Cognition Blog, <https://cognition.ai/blog/swe-grep#rl-for-training-fast-parallel-agentic-code-search-models>, October 2025. Accessed: 2026-05-05.

- Pew Research Center. Google users are less likely to click on links when an ai summary appears in the results. <https://www.pewresearch.org/short-reads/2025/07/22/google-users-are-less-likely-to-click-on-links-when-an-ai-summary-appears-in-the-results/>, July 2025. Accessed 2026-01-26.
- Stephen Robertson and Hugo Zaragoza. *The probabilistic relevance framework: BM25 and beyond*, volume 4. Now Publishers Inc, 2009.
- Stephen E Robertson. The probability ranking principle in ir. *Journal of documentation*, 33(4):294–304, 1977.
- Anshumali Shrivastava. Understanding the fundamental limitations of vector-based retrieval for building llm-powered chatbots— (part 1/3). Medium, <https://medium.com/thirdai-blog/understanding-the-fundamental-limitations-of-vector-based-retrieval-for-building-llm-powered-48bb7b5a57b3>, June 2023. Accessed: 2026-05-05.
- Anshumali Shrivastava. Deepmind calls out embedding limits: Why single-vector retrieval falls short — an attention perspective. Medium, https://medium.com/@Anshumali_/deepmind-calls-out-embedding-limits-why-single-vector-retrieval-falls-short-an-attention-2a930d477d80, September 2025. Accessed: 2026-05-05.
- Shreyas Subramanian, Adewale Akinfaderin, Yanyan Zhang, Ishan Singh, Mani Khanuja, Sandeep Singh, and Maira Ladeira Tanke. Keyword search is all you need: Achieving rag-level performance without vector databases using agentic tool use. *arXiv preprint arXiv:2602.23368*, 2025.
- Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogenous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663*, 2021.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10014–10037, Toronto, Canada, July 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.acl-long.557. <https://aclanthology.org/2023.acl-long.557/>.
- Baoyi Wang, Xingliang Wang, Guochang Li, Chen Zhi, Junxiao Han, Xinkui Zhao, Nan Wang, Shuiguang Deng, and Jianwei Yin. Greprag: An empirical study and optimization of grep-like retrieval for code completion. *arXiv preprint arXiv:2601.23254*, 2026.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*, 2022.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022.
- Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. Browsecomp: A simple yet challenging benchmark for browsing agents. *arXiv preprint arXiv:2504.12516*, 2025.
- Orion Weller, Michael Boratko, Iftekhar Naim, and Jinhyuk Lee. On the theoretical limitations of embedding-based retrieval. *arXiv preprint arXiv:2508.21038*, 2025.
- Peilin Wu, Mian Zhang, Kun Wan, Wentian Zhao, Kaiyu He, Xinya Du, and Zhiyu Chen. Hiprag: hierarchical process rewards for efficient agentic retrieval augmented generation. *arXiv preprint arXiv:2510.07794*, 2025.
- Yutao Xie, Nathaniel Thomas, Nicklas Hansen, Yang Fu, Li Erran Li, and Xiaolong Wang. TIPS: Turn-level information-potential reward shaping for search-augmented LLMs. In *The Fourteenth International Conference on Learning Representations*, 2026. <https://openreview.net/forum?id=eBMOr6a84z>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822, 2023.
- Fengji Zhang, Xinyao Niu, Chengyang Ying, Guancheng Lin, Zhongkai Hao, Zhou Fan, Chengen Huang, Jacky Keung, Bei Chen, and Junyang Lin. A²search: Ambiguity-aware question answering with reinforcement learning. *arXiv preprint arXiv:2510.07958*, 2025.
- Shengjun Zhang, Zhang Zhang, Chensheng Dai, and Yueqi Duan. E-grpo: High entropy steps drive effective reinforcement learning for flow models. *arXiv preprint arXiv:2601.00423*, 2026.

Tiancheng Zhao, Xiaopeng Lu, and Kyusong Lee. Sparta: Efficient open-domain question answering via sparse transformer matching retrieval. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 565–575, 2021.

Appendix

A BrowseComp-Wikipedia Search Prompt

For the BrowseComp-Wikipedia experiment (Section 4.4), SIRA is driven by the single prompt shown in Figure 4. The prompt instructs the LLM to first solve the clue chain privately, validate candidate Wikipedia categories through the `search_categories` tool, and then commit to one or more grounded `bm25_search` calls under a fixed 100-unique-page retrieval budget. The template variable `{problem}` is replaced by the BrowseComp question at evaluation time.

You are a budgeted Wikipedia BM25 search agent for hard clue-chain questions. Your goal is to put the gold page in the final top 10 or top 100 using at most 100 unique BM25-retrieved pages.

TOOLS:

`search_categories(query: str, k: int = 20) -> str`

Search real Wikipedia category names. Use this BEFORE `bm25_search` to find closest real category anchors for your intended entity/type/geography facet. If the exact imagined category is absent, broaden the query and choose the closest semantically aligned real category returned by the tool.

`bm25_search(keywords: str, categories: list[str] = [], k: int = 10) -> str`

Runs BM25 and spends the fixed 100-unique-page retrieval budget. Returns only new pages. Call only when keywords are concrete and categories are real/validated. Keep issuing targeted BM25 calls until this returns `BUDGET_EXHAUSTED`.

`fetch_page(page_id: int) -> str`

Fetch title + snippet for a page already returned by `bm25_search`. Use this only to rank or verify candidates already inside the 100-page budget.

WORKFLOW:

1. Privately solve the puzzle as far as possible. Identify the likely answer type, answer hypotheses, clue entities, bridge entities, locations, dates, works, organizations, and relationships.

2. Before every `bm25_search`, validate categories. Call `search_categories` with broad and narrow variants until you find the closest real category anchors for that query facet. Do not invent categories. Do not leave categories empty merely because the exact category is missing; pick the closest real semantically aligned category returned by the tool.

3. Gate each BM25 query. Fire `bm25_search` only if the keywords contain concrete Wikipedia-surface strings and the categories are real and aligned with the same hypothesis/facet. If categories are broad but aligned, use them with strong rare keywords. If keywords are generic or the category points to a different domain, do not spend budget on that query.

4. Use multiple targeted BM25 queries until the 100-page budget is exhausted. Do not stop merely because you found a plausible candidate; continue spending the budget on distinct high-confidence facets so the final 100 pages have the best possible recall. Good query facets include: final answer hypothesis, bridge entity, clue entity, event/work/institution facet, and category/type/geography facet.

5. Track all `page_ids` returned by `bm25_search`. Optionally fetch the most promising candidates to rank them, but keep querying until `bm25_search` returns `BUDGET_EXHAUSTED` or `unique_pages_seen` reaches 100. Final `page_ids` must come from pages actually seen in `bm25_search`/`fetch_page` results.

FINAL OUTPUT:

Output ONLY JSON with exactly these fields:

"page_ids" — ordered list of up to 100 `page_ids`, most likely gold first. Use only `page_ids` seen in this session.

"rationale" — one short paragraph explaining the ranking evidence.

"hypothesis" — one compact sentence naming the likely answer and key clue entities.

Do not output markdown fences or prose outside JSON.

Question:

`{problem}`

Figure 4 Prompt used to drive SIRA on the BrowseComp-Wikipedia benchmark. The LLM reasons privately, grounds candidate Wikipedia categories through the `search_categories` tool, and issues grounded `bm25_search` calls under a fixed 100-unique-page budget. The variable `{problem}` is replaced by the BrowseComp question.

B BrowseComp-Wikipedia Case Study: SIRA vs. Perplexity

Figure 5 shows the complete tool-call traces for a single BrowseComp-Wikipedia query (gold page: *LocoRoco*). SIRA reaches the gold page at rank 1 after a handful of category-grounded BM25 calls, while the agentic Perplexity-search baseline issues 72 distinct searches, never retrieves the gold page, and commits to an incorrect answer.

Question (paraphrased to avoid benchmark leakage; gold page: LocoRoco, page id 4905353):

A certain video game made its debut on a gaming platform in July of a year somewhere from 2004 through 2007. The person who created it was born in April of a year in the 1970–1973 range and drew the game’s earliest sketches at some point between 2002 and 2005. One of the title’s music composers was born in June of a year from 1968 to 1971. Which game is being described?

SIRA (ours) — gold retrieved at rank 1. 11 `bm25_search` calls, 9 `search_categories` calls, 9 `fetch_page` calls; budget not exhausted. Trace (* = call whose results contained the gold page):

0. [search_categories] query: video games released July 2005 2006
1. [search_categories] query: Japanese video game designers born 1970s
2. [search_categories] query: video games 2005 2006 2007
3. [search_categories] query: indie video games
- * 4. [bm25_search] keywords: LocoRoco game PSP 2006 | categories: ['2006 video games'] | k=5
5. [search_categories] query: PlayStation Portable games
6. [fetch_page] page_id: 4905353
7. [bm25_search] keywords: Tsutomu Kouno LocoRoco designer creator | categories: ['Japanese video game designers'] | k=5
8. [search_categories] query: video game composers born 1968 1969 1970
9. [search_categories] query: video game composers
10. [search_categories] query: video game music composers Japanese
11. [bm25_search] keywords: Fumito Ueda game designer born April 1970 | categories: ['Japanese video game designers'] | k=5
12. [fetch_page] page_id: 3041309
13. [bm25_search] keywords: video game released July 2005 sketches development | categories: ['2005 video games'] | k=10
14. [bm25_search] keywords: Kow Otani composer born June | categories: ['Japanese video game composers'] | k=5
15. [search_categories] query: Shadow of the Colossus
16. [fetch_page] page_id: 3402872
17. [bm25_search] keywords: Osu Tatakae Ouendan game July 2005 | categories: ['2005 video games'] | k=5
18. [fetch_page] page_id: 2740521
19. [bm25_search] keywords: Keiichi Yano video game designer iNiS born | categories: ['Japanese video game designers'] | k=5
20. [fetch_page] page_id: 20435930
21. [bm25_search] keywords: game designer born April 1971 creator sketches | categories: ['Japanese video game designers'] | k=5
22. [bm25_search] keywords: game designer born April 1972 creator | categories: ['Japanese video game designers'] | k=5
23. [fetch_page] page_id: 328218
24. [fetch_page] page_id: 5262474
25. [bm25_search] keywords: Shinta Nojiri Metal Gear Acid game director July | categories: ['2005 video games', '2006 video games'] | k=5
26. [fetch_page] page_id: 851696
27. [fetch_page] page_id: 1918270
28. [bm25_search] keywords: Tsutomu Kouno LocoRoco creator born | categories: ['Japanese video game designers'] | k=5

Perplexity — 72 searches, gold page never retrieved; final answer incorrect. Query trace:

1. video game creator born April 1970 1971 1972 1973 first sketches 2002 2003 2004 2005
2. video game debuted July 2004 2005 2006 2007 platform
3. Painkiller video game 2004 release date platform July
4. Adrian Chmielarz Painkiller game development sketches history
5. Painkiller video game Xbox release date July 2006 music composer
6. Painkiller video game soundtrack composer born June 1968 1969 1970 1971

7. Shoji Meguro video game composer games list Persona
8. Painkiller video game full Wikipedia article music composer development
9. Persona 3 video game release date July 2006 director creator
10. Trauma Center Under the Knife release date July 2004 2005 director creator born April
11. Katsura Hashino born birthday Persona 3 director development history
12. Persona 3 development history first sketches concept 2002 2003 2004
13. Katsura Hashino birthday born date April 1970 1971 1972 1973
14. Persona 3 Wikipedia full article director producer composer
15. Katsura Hashino Atlus director born 1970s April
16. Persona 3 game creator director Hashino birth date Wikipedia
17. video game released July 2004 2005 2006 2007 creator born April 1970-1973 composer born June 1968-1971
18. Maken X Atlus 1999 director Katsura Hashino born April
19. Persona 3 Wikipedia infobox director producer release date
20. Todd Howard born 1970 April game director Bethesda first sketches 2002 2003 2004 2005
21. The Elder Scrolls IV Oblivion release date July 2006 development first sketches
22. Todd Howard born April 1970 birthday Wikipedia
23. video game composer born June 1968 1969 1970 1971 games released July 2004 2005 2006 2007
24. Persona 3 development "first sketches" OR "concept art" OR "early development" 2003 2004
25. killer7 video game July 2005 director creator Suda51 born April
26. Suda51 Goichi Suda born April 1968-1973 game director
27. killer7 music composer born June 1968 1969 1970 1971
28. Hidetaka Suehiro SWERY games directed 2004 2005 2006 2007 release July
29. The Idolmaster video game 2005 arcade director creator born April composer born June
30. Persona 3 Wikipedia "first sketches" OR "initial sketches" development 2003
31. Akihiro Ishihara Idolmaster director born April 1970 1971 1972 1973
32. video game Wikipedia "first sketches" development creator director born April
33. The Idolmaster video game Wikipedia development history creator Ishihara first sketches
34. Jason Jones Bungie game director Halo released July 2004 2005 2006 2007
35. game released July 2004-2007 creator born April 1970-1973 "first sketches"
36. Katsura Hashino Wikipedia personal life born 1970 1971 1972 1973
37. Michel Ancel born April 1972 Beyond Good Evil game development sketches
38. video game Wikipedia "first sketches" development 2002 2003 2004 2005 released July
39. video game Wikipedia "first sketches" 2002 2003 2004 2005 creator born April
40. Shantae game creator Matt Bozon born April 1970 1971 1972 1973 first sketches
41. Viva Piñata video game release date July 2006 Gregg Mayles development first sketches
42. Gregg Mayles Rare games 2004 2005 2006 2007 director creator Viva Pinata
43. Grant Kirkhope born June 1968 1969 1970 1971 composer birthday
44. Viva Piñata video game release date November 2006 Xbox 360
45. video game composer born June 1968 games released July 2004 2005 2006 2007
46. Persona 3 Wikipedia full article infobox director producer release date July 13 2006
47. Motoi Sakuraba born June 1968 1969 1970 1971 composer Tales of Symphonia
48. Tales of Symphonia director creator born April 1970-1973 development first sketches
49. Prey 2006 video game director creator born April 1970-1973 development first sketches composer
50. Persona 3 Wikipedia Katsura Hashino born April 1970 1971 1972 1973
51. Ōkami video game release date July 2006 director Hideki Kamiya born April composer
52. Hideki Kamiya born April 1970 1971 1972 1973 birthday
53. Shin Megami Tensei Digital Devil Saga release date July 2004 director Katsura Hashino
54. video game released July 2004-2007 composer born June 1968-1971 creator born April 1970-1973
55. Shin Megami Tensei Digital Devil Saga development history first sketches 2002 2003 Wikipedia
56. Shin Megami Tensei Digital Devil Saga Wikipedia full article development composer music
57. Katsura Hashino Atlus director "born" OR "birthday" OR "birth date" April
58. Persona 3 Wikipedia "first sketches" OR "initial concept" OR "concept art" development 2003
59. 2007 in video games July releases list Wikipedia
60. 2004 in video games July releases list Wikipedia
61. Painkiller Xbox port release date 2006 exact date July
62. Painkiller video game music composer born June 1968 1969 1970 1971
63. Painkiller Hell Wars Xbox release date 2006 exact month
64. Painkiller video game 2004 music composer soundtrack Wikipedia
65. Painkiller video game 2004 music composer Marcin Przybyłowicz OR Tomasz Hajduk born June
66. Painkiller video game People Can Fly development history first sketches 2002 2003 Adrian Chmielarz
67. Painkiller video game Wikipedia full article development music sound
68. Painkiller 2004 video game soundtrack composer Tomasz Hajduk OR Marcin Przybyłowicz OR Pawel Blaszcak
69. Painkiller 2004 game music composer Tomasz Hajduk Wikipedia
70. Adrian Chmielarz Painkiller "first sketches" OR "concept" OR "development began" 2002
71. Painkiller video game 2004 Wikipedia "composer" OR "music" OR "soundtrack" Tomasz Hajduk
72. Katsura Hashino Wikipedia "born" April 1970 1971 1972 1973 Atlus director

Perplexity final answer (incorrect): fixates on Katsura Hashino / Persona 3 and Painkiller, concludes that Wikipedia "doesn't

have Katsura Hashino's birth date," and never identifies *LocoRoco* as the answer.

Figure 5 Full tool-call traces for one BrowseComp-Wikipedia query (gold page: *LocoRoco*); the question is paraphrased to avoid benchmark leakage. SIRA grounds the indirect clues into category-anchored BM25 calls and retrieves the gold page at rank 1 in 11 retrieval calls (the winning call is marked ★). The agentic Perplexity-search baseline issues 72 distinct web searches, cycles through incorrect candidates (Persona 3, Painkiller), never surfaces the gold page, and commits to a wrong answer.